RACE
Regular Analysis Through Continuous Exercise

Newton's Laws of Motion : RACE-01

1. In order to jump off the floor, the floor must exert a force on you
(A) in the direction of and equal to your weight.
(B) opposite to and equal to your weight.
(C) opposite to and less than your weight.
(D) opposite to and greater than your weight.

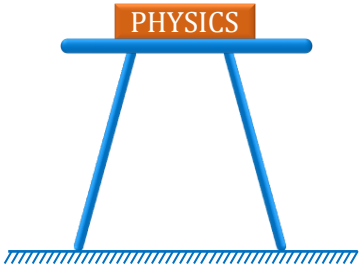
PNL267

2. **Statement 1 :** When a block is suspended from a fixed support with a light string, the weight of the block is equal in magnitude to the force string applies on the support.
And
Statement 2 : When a body applied force on another body, the other body applies an equal and opposite force on the first body.
(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

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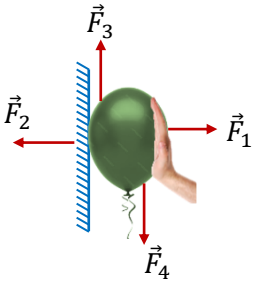
3. A book is lying on the table. What is the angle between the action of the book on the table and the reaction of the table on the book?



(A) 0 rad (B) $\pi/2$ rad (C) $2\pi/3$ rad (D) π rad

PNL269

4. In the figure shown, a balloon is pressed against a wall. It is in equilibrium and maximum compressed state. $\vec{F}_1$ = force of balloon on hand of man ; $\vec{F}_2$ = force of balloon on wall ; $\vec{F}_3$ = friction; $\vec{F}_4$ = weight of balloon. Choose the correct statement.

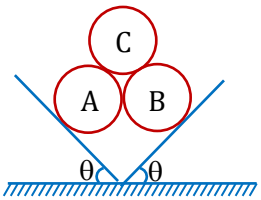


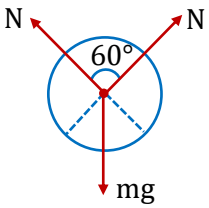
(A) $\vec{F}_1$ and $\vec{F}_2$ are action reaction pairs.
(B) $\vec{F}_3$ and $\vec{F}_4$ are action reaction pairs.
(C) $\vec{F}_2$ and $\vec{F}_3$ are action reaction pairs.
(D) $\vec{F}_4$ and gravitational force exerted by balloon on earth are action reaction pairs.

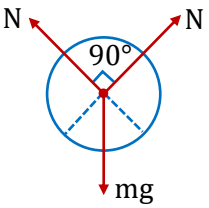
PNL270

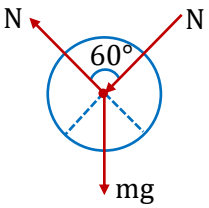
For More Join: @IITJEE_Advanced

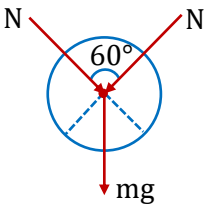
5. *A*, *B* and *C* are three identical smooth sphere placed on frictionless inclined plane as shown in figure then *F.B.D.* of sphere *C* is :



(A) 

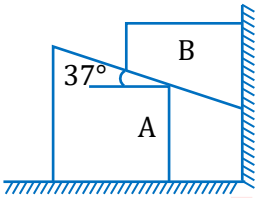
(B) 

(C) 

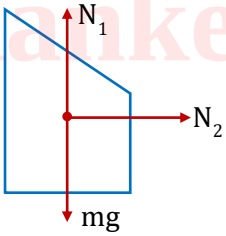
(D) 

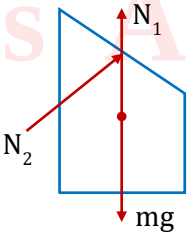
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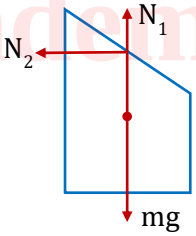
6. Two blocks *A* and *B* are placed as shown in figure. There is no friction anywhere then *F.B.D.* of block *A*.

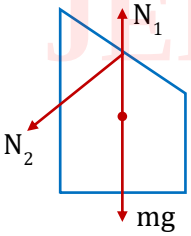


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(A) 

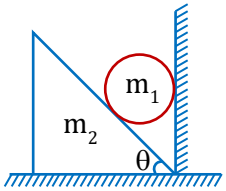
(B) 

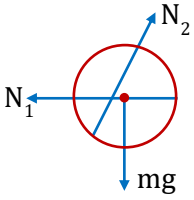
(C) 

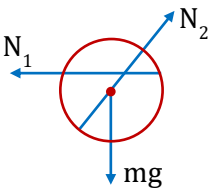
(D) 

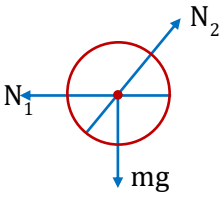
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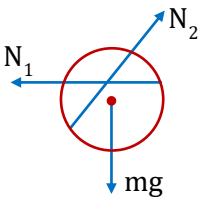
7. Two mass m_1 and m_2 are placed as shown in figure. Assume all surface are frictionless then the correct *F.B.D.* of m_1 is:



(A) 

(B) 

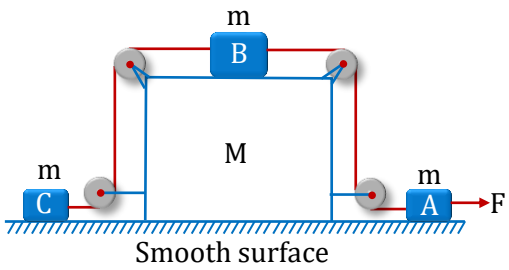
(C) 

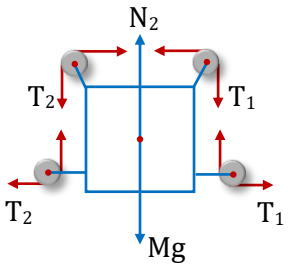
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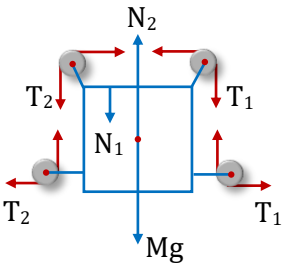
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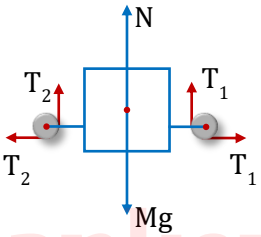
For More Join: @IITJEE_Advanced

8. Block *A*, *B* and *C* connected with light string with larger block *M* as shown in figure. Pullies are light and frictionless. There is no friction anywhere. Which is the correct *F.B.D.* of larger block of mass *M*.



(A) 

(B) 

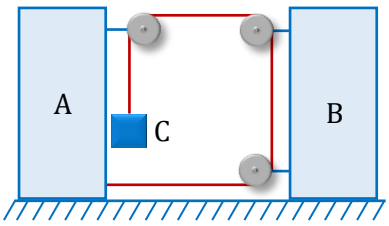
(C) 

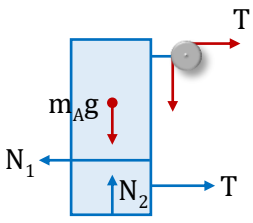
(D) None of these

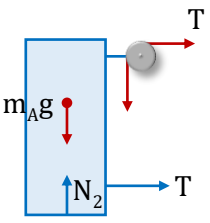
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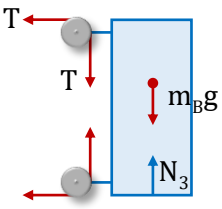
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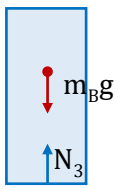
9. Block *A* and *B* are connected by light string as shown in figure. All surface are frictionless then which is the correct *F.B.D.* of block *B* and block *A*.



(A) 

(B) 

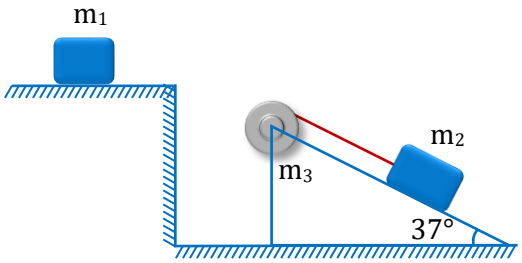
(C) 

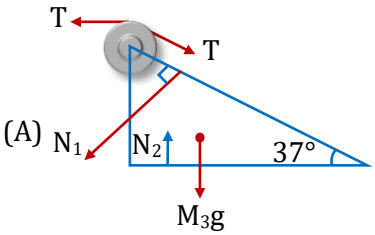
(D) 

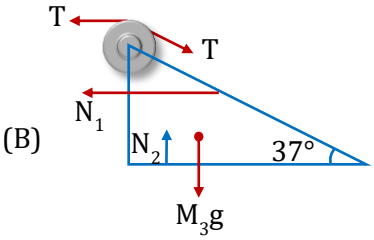
PNL275

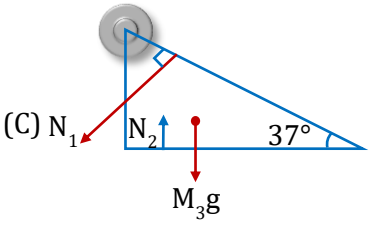
For More Join: @IITJEE_Advanced

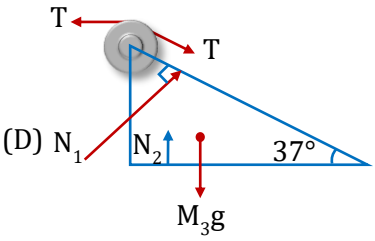
10. Masses m_1, m_2 and m_3 are connected light string as shown in figure. There is no friction at anywhere. Which is the *F.B.D.* of mass m_3.



(A)


(B)


(C)


(D)


PNL276

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	D	D	A	D	C	B	AC	A

For More Join: @IITJEE_Advanced

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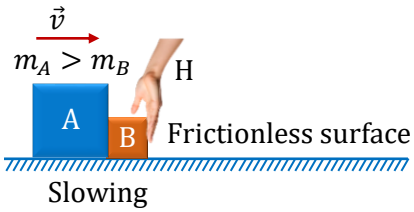
Newton's Laws of Motion : RACE-02

1. Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ are acting on a particle of mass m such that $\vec{F}_1+\vec{F}_2+\vec{F}_3=\vec{0}$. If the force $\vec{F}_1$ is now removed then the acceleration of the particle is:

(A) $(\vec{F}_2-\vec{F}_3)\,/\,m$ (B) $\vec{F}_2/m$ (C) $\vec{F}_1/m$ (D) $-\vec{F}_1/m$

PNL277

2. Boxes A and B are sliding to the right across a frictionless table. The hand H is slowing them down. The mass of A is larger than the mass of B. Here $F_{A\text{ on }B}$ represents force on B due to A and same for other:



(A) $F_{B\text{ on }H}=F_{H\text{ on }B}=F_{A\text{ on }B}=F_{B\text{ on }A}$

(B) $F_{B\text{ on }H}=F_{H\text{ on }B}>F_{A\text{ on }B}=F_{B\text{ on }A}$

(C) $F_{B\text{ on }H}=F_{H\text{ on }B}<F_{A\text{ on }B}=F_{B\text{ on }A}$

(D) $F_{B\text{ on }H}=F_{H\text{ on }B}>F_{A\text{ on }B}>F_{B\text{ on }A}$

PNL278

3. Consider the following statement about the blocks shown in the diagram that are being pushed by a constant force on a frictionless table.



(i) All blocks move with the same acceleration.

(ii) The net force on each block is the same.

Which of these statements are/is correct?

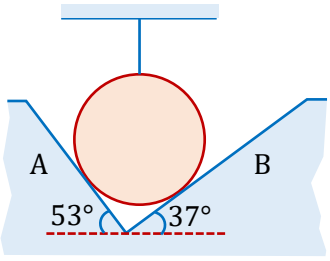
(A) (i) Only (B) (ii) Only (C) Both (i) and (ii) (D) Neither (i) nor (ii)

PNL279

4. A boy pushes a box of mass 2 kg with a force $\vec{F}=(20\hat{\imath}+10\hat{\jmath})N$ on a frictionless surface. If the box was initially at rest, then _____ m is displacement along the x-axis after 10 s.

PNL280

5. In the system is shown a 10 kg horizontal cylinder suspended from string from the ceiling and supported by two fixed frictionless slopes A and B.



If normal reaction applied by slope A on the cylinder is 30 N, which of the following conclusion can you make? [g = 10 m/s^2]

(A) No tension force is applied by the string on the cylinder.

(B) Tension force applied by the string on the cylinder is 50 N.

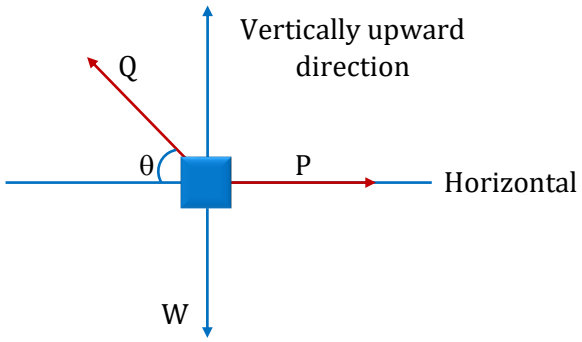
(C) Normal reaction applied by the slope B on the cylinder is 60 N.

(D) Normal reaction applied by the slope B on the cylinder is 40 N.

PNL281

For More Join: @IITJEE_Advanced

6. A block of weight W is kept in equilibrium by two forces P and Q as shown in the figure. Which of the following give the condition for the block to be in equilibrium?

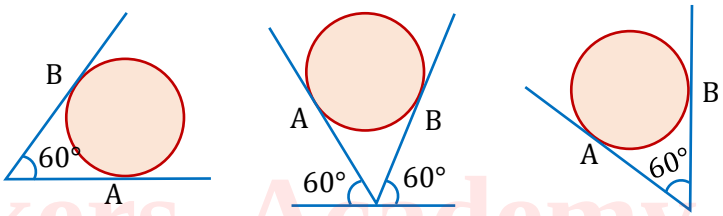


(A) $W = P\tan\theta$ (B) $W = P\cot\theta$

(C) $W = Q\sin\theta$ and $P = Q\cos\theta$ (D) $P = Q\sin\theta$ and $W = Q\cos\theta$

PNL282

7. An iron sphere weighing 10 N rests in a V shaped smooth trough whose sides form an angle of 60° as shown in the figure. Then the reaction forces are



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(A) $R_A = 10\ N$ and $R_B = 0$ in case (i) (B) $R_A = 10\ N$ and $R_B = 10\ N$ in case (ii)

(C) $R_A = \frac{20}{\sqrt{3}}\ N$ and $R_B = \frac{10}{\sqrt{3}}\ N$ in case (iii) (D) $R_A = 10\ N$ and $R_B = 10\ N$ in all the three cases

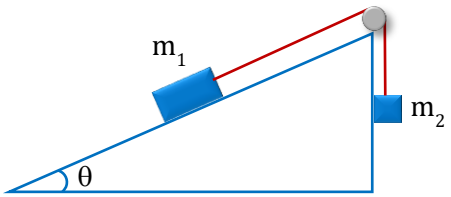
PNL283

8. A mass of 10 kg is suspended vertically by a rope from the roof. When a horizontal force is applied on the rope at some point, the rope deviated at an angle of 45° at the roof point. If the suspended mass is at equilibrium, the magnitude of the force applied is ($g = 10\ ms^{-2}$)

(A) 200 N (B) 100 N (C) 140 N (D) 70 N

PNL284

9. Two bodies of masses $m_1 = 5\ kg$ and $m_2 = 3\ kg$ are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be : [Take $g = 10\ ms^{-2}$]

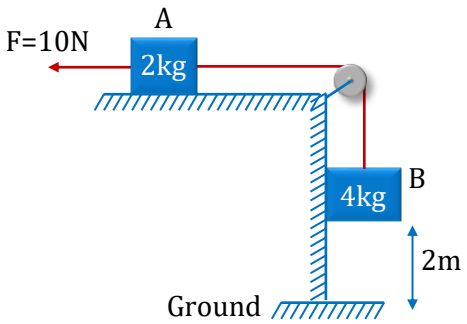


(A) 30 N (B) 40 N (C) 50 N (D) 60 N

PNL285

For More Join: @IITJEE_Advanced

10. The system shown in the diagram is released from rest. Pulley and string are massless. Neglect friction everywhere. (g = 10 ms^{-2})



(A) Tension in the string during motion is 20N

(B) Acceleration of blocks during motion is 5 ms^{-2}

(C) Block B reaches ground in $2/\sqrt{5}s$

(D) All of the above statements are incorrect

PNL286

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ANSWER KEY

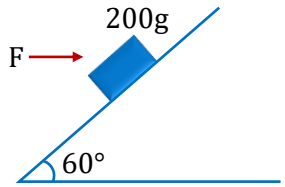
Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	A	500	BD	C	ABC	B	B	ABC

For More Join: @IITJEE_Advanced

RACE
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Newton's Laws of Motion : RACE-03

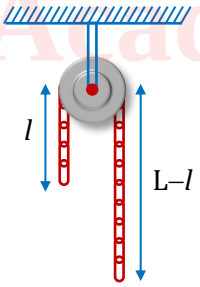
1. A block of mass 200 g is kept stationary on a smooth inclined plane by applying a minimum horizontal force $F = \sqrt{x}N$ as shown in figure. The value of x = _____.



PNL287

2. A uniform metal chain of mass m and length 'L' passes over a massless and frictionless pulley. It is released from rest with a part of its length 'l' is hanging on one side and rest of its length '$L - l$' is hanging on the other side of the pulley. At a certain point of time, when $l = \frac{L}{x}$, the acceleration of the chain is $\frac{g}{2}$. The value of x is _____.

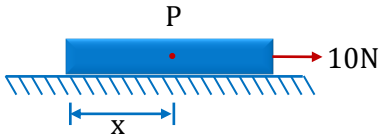
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(A) 6 (B) 2 (C) 1.5 (D) 4

PNL288

3. A rope whose mass per unit length varies according to the relation $\lambda = 4x$ $[kg/m]$ is being pulled by a constant horizontal force of $10N$ over a smooth floor (length of rope, $L = 1m$) as shown in the figure. Find:



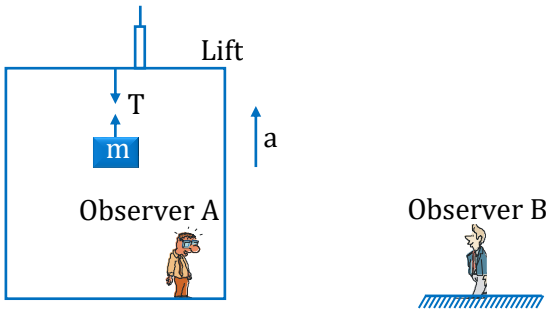
(a) acceleration of the rope

(b) tension at point P if $x = \frac{L}{4}$

PNL289

For More Join: @IITJEE_Advanced

4. A block is hanging by a rope in a lift, and lift is accelerating upward. This motion is observes by two observer *A* and B. Here observer '*A*' is in lift and observer '*B*' is at rest on ground, then :-



(A) Tension in rope is same with respect to observer *A* and observer *B*.

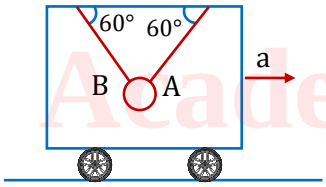
(B) Tension in rope is $m(g + a)$ w.r.t observer '*B*' and tension is Mg w.r.t. observer *A*.

(C) Tension is rope is $M(g + a)$ w.r.t. observer *B* and zero w.r.t. observer *A*.

(D) Net force on the block is zero w.r.t. observer *A*.

PNL290

5. A steel ball is suspended from the ceiling of an accelerating carriage by means of two cords *A* and *B*. Determine the acceleration a of the carriage which will cause the tension in *A* to be twice that in *B*.



Rankers Academy JEE

PNL291

6. A person standing on a spring balance inside a stationary lift measures 60 kg. The weight of that person if the lift descends with uniform downward acceleration of 1.8 m/s^2 will be____ N.
[$g = 10\ m/s^2$]

PNL292

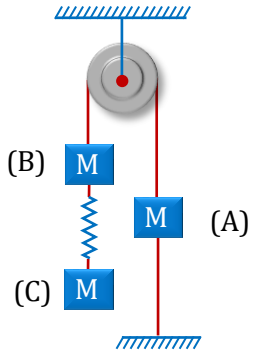
7. In the given figure choose the correct statement regarding the acceleration of masses just after the string between '*A*' and ground is cut.

(A) All the three blocks move with same acceleration $\left(\frac{g}{3}\right)$.

(B) Acceleration of $a_C = a_B = 0$ and $a_A = \frac{g}{2}$.

(C) $a_A = a_B = \frac{g}{2}$; $a_C = 0$.

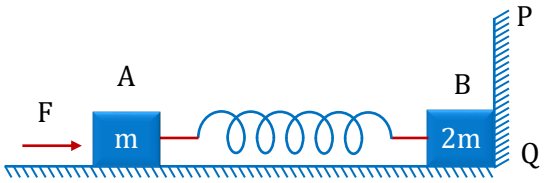
(D) $a_A = a_B = \frac{g}{3}$; $a_C = 0$.



PNL293

For More Join: @IITJEE_Advanced

8. Figure shows a system of two blocks connected by a spring, kept on a frictionless surface. Block *A* is in equilibrium. Suddenly supporting vertical wall *PQ* is removed, then which of the following statements is/are correct :-



(A) Initial acceleration of block *A* is $\frac{F}{m}$

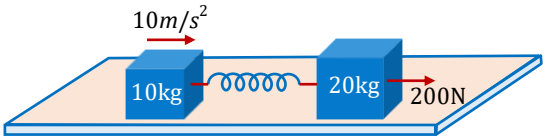
(B) Initial acceleration of block *B* is $\frac{F}{2m}$

(C) Initial acceleration of block *B* is zero

(D) Initial acceleration of block *A* is $\frac{F}{3m}$

PNL294

9. The masses of 10 *kg* and 20 *kg* respectively are connected by a massless spring as shown in figure. A force of 200 *N* acts on the 20 *kg* mass. At the instant shown, the 10 *kg* mass has acceleration 10 m/sec^2 then calculate the acceleration of 20 *kg* mass in m/s^2.



PNL295

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ANSWER KEY

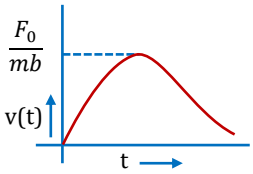
Question	1	2	3	4	5	6	7	8	9
Answer	12	D	(a) 5 m/s^2, (b) 5/8 *N*	AD	$\frac{g}{3\sqrt{3}}$	492	C	B	5

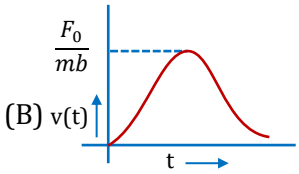
For More Join: @IITJEE_Advanced

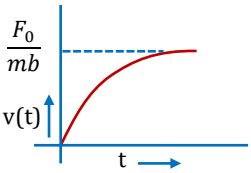
RACE
Regular Analysis Through Continuous Exercise

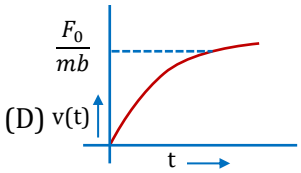
Newton's Laws of Motion : RACE-04

1. A particle of mass m is at rest at the origin at time $t = 0$. It is subjected to a force $F(t) = F_0e^{-bt}$ in the x direction. Its speed $v(t)$ is depicted by which of the following curves ?

(A) 

(B) 

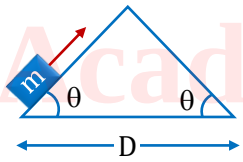
(C) 

(D) 

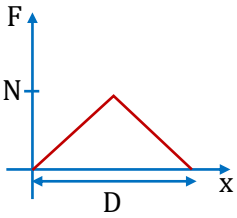
PNL296

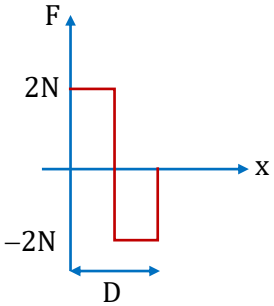
2. An object of mass 'm' is being moved with a constant velocity under the action of an applied force of $2N$ along a frictionless surface with following surface profile.

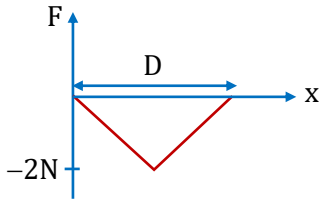
Rankers Academy JEE

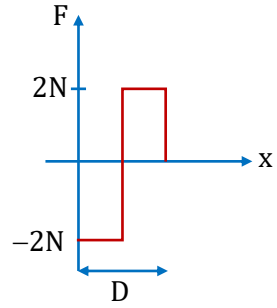


The correct applied force vs distance graph will be:

(A) 

(B) 

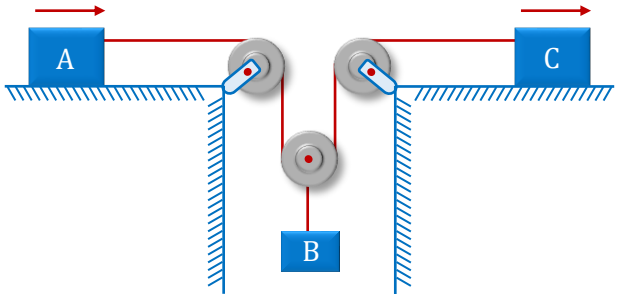
(C) 

(D) 

PNL297

For More Join: @IITJEE_Advanced

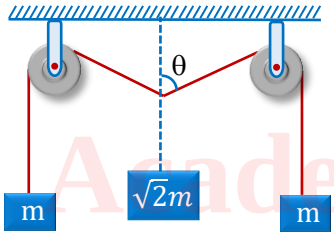
3. Block A and C start from rest and move to the right with acceleration $a_A = 12t\ m/s^2$. And $a_c = 3\ m/s^2$. Here t is in seconds. The time when block B again comes to rest is



(A) $2s$ (B) $1s$ (C) $3/2s$ (D) $1/4s$

PNL299

4. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be

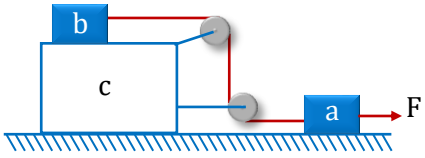


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(A) 0° (B) 30° (C) 45° (D) 60°

PNL300

5. Three blocks a, b and c of masses 10 kg, 10 kg and 20 kg respectively are arranged as shown in figure. All the surfaces are frictionless and string is inextensible. A constant force $F = 20\ N$ is applied on block 'a' as shown. Pulleys and string are light. Part of the string connecting both pulleys is vertical and part of the strings connecting pulleys with masses a and b are horizontal.



(A) Acceleration of blocks a, b and c is 0.5 m/s^2

(B) Acceleration of block b is 1 m/s^2

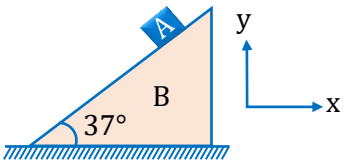
(C) Tension in the string is 10 N

(D) Acceleration of block c is 0.5 m/s^2

PNL301

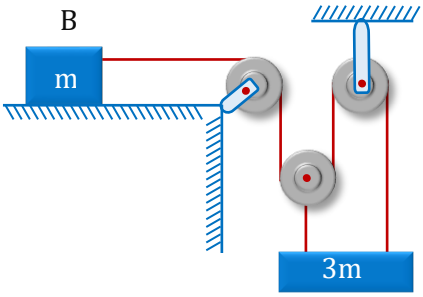
For More Join: @IITJEE_Advanced

6. In the figure shown the acceleration of A is, $\vec{a}_A = (15\hat{\imath} + 15\hat{\jmath})\ m/s^2$. If A is sliding on B then the acceleration of B is $p\,\hat{\imath}\ m/s^2$, then find the value of p.



PNL302

7. Find out the acceleration of the block B in the following systems :



PNL328

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ANSWER KEY

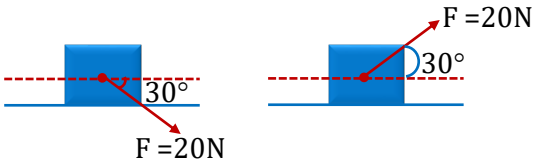
Question	1	2	3	4	5	6	7
Answer	D	B	D	C	BC	–5	3g/4

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Friction : RACE-05

1. A block of mass 5 *kg* is (i) pushed in case (A) and (ii) pulled in case (B), by a force $F = 20$ *N*, making an angle of 30° with the horizontal, as shown in the figures. The coefficient of friction between the block and floor is $\mu = 0.2$. The difference between the accelerations of the block, in case (B) and case (A) will be : ($g = 10\ ms^{-2}$)



(A) (B)

(A) 0 ms^{-2} (B) 0.8 ms^{-2} (C) 0.4 ms^{-2} (D) 3.2 ms^{-2}

PNL304

2. A block of mass 40 *kg* slides over a surface, when a mass of 4 *kg* is suspended through an inextensible massless string passing over frictionless pulley as shown below. The coefficient of kinetic friction between the surface and block is 0.02. The acceleration of block is.
(Given $g = 10\ ms^{-2}$.)



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(A) 1 ms^{-2} (B) 1/5 ms^{-2} (C) 4/5 ms^{-2} (D) 8/11 ms^{-2}

PNL305

3. A block of mass 10 *kg* starts sliding on a surface with an initial velocity of 9.8 ms^{-1}. The coefficient of friction between the surface and bock is 0.5. The distance covered by the block before coming to rest is : [Use $g = 9.8\ ms^{-2}$]
(A) 4.9 *m* (B) 9.8 *m* (C) 12.5 *m* (D) 19.6 *m*

PNL306

4. A block of mass 1 *kg* is found to be moving with velocity $10\hat{j}$ *m*/sec on rough horizontal surface (*xy*-plane) ($\mu = 0.2$) which is also moving with velocity $10\hat{i}$*m*/sec. The smallest angle made by friction acting on block with positive x-axis is given by $\alpha\ \pi$. Fill the value of 12 α in OMR sheet.

PNL307

5. A block of mass 1 *kg* lies on a horizontal surface in a truck. The coefficient of static friction between the block and the surface is 0.6 If the acceleration of the truck is 5 m/s^2, the frictional force acting on the block is newton.

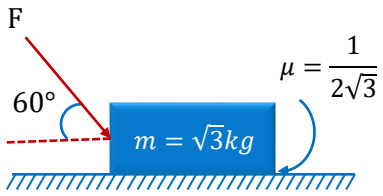
PNL308

6. A block of mass 2 *kg* is pushed against a rough vertical wall with a force of 30 *N*, coefficient of static friction being 0.5. Another horizontal force of 15 *N* is applied on the block in a direction parallel to the wall. What is the acceleration of block (in m/s^2) ?

PNL309

For More Join: @IITJEE_Advanced

7. What is the maximum value of the force F (in newton) such that the block shown in the arrangement, does not move :



PNL310

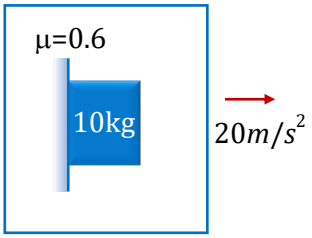
8. Car is accelerating with acceleration = 20 m/s^2. A box of mass m = 10 kg that is placed inside the car, it is put in contact with the vertical wall of car as shown. The friction coefficient between the box and the wall is μ = 0.6.

(A) The acceleration of the box will be 20 m/sec^2.

(B) The friction force acting on the box will be 100 N.

(C) The contact force between the vertical wall and the box will be $100\sqrt{5}N$.

(D) The net contact force between the vertical wall and the box is only of electromagnetic in nature.



PNL311

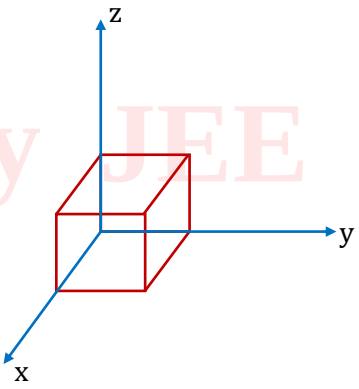
9. A solid cube of mass 5 kg is placed on a rough horizontal surface, in xy-plane as shown. The friction coefficient between the surface and the cube is 0.4. An external force $\vec{F} = 6\hat{\imath} + 8\hat{\jmath} + 20\hat{k}$N is applied on the cube. (Use g = 10 m/s^2)

(A) The block starts slipping over the surface.

(B) The friction force on the cube by the surface is 10 N.

(C) The friction force acts in xy-plane at angle 127° with the positive x-axis in clockwise direction.

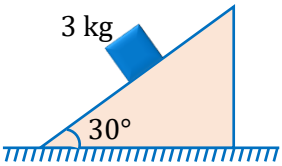
(D) The contact force exerted by the surface on the cube is $10\sqrt{10}N$.



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PNL312

10. A block of mass 3 kg is at rest on a rough inclined plane as shown in the figure. The magnitude of net force exerted by the surface on the block will be (g = 10 m/s^2)



(A) 26 N (B) 19.5 N (C) 10 N (D) 30 N

PNL313

ANSWER KEY

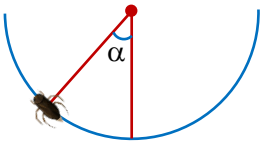
Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	B	3	5N	5	20	ABCD	BCD	D

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Friction : RACE-06

1. An insect crawls up a hemispherical surface very slowly. The coefficient of friction between the insect and the surface is 1/3. If the line joining the center of the hemispherical surface to the insect makes an angle α with the vertical the maximum possible value of α is given by



(A) $\cot\alpha = 3$ (B) $\tan\alpha = 3$ (C) $\sec\alpha = 3$ (D) $\operatorname{cosec}\alpha = 3$

PNL314

2. A body of mass 'm' is launched up on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of friction between the body and plane is $\frac{\sqrt{x}}{5}$. If the time of ascent is half of the time of descent. The value of x is ____.

PNL315

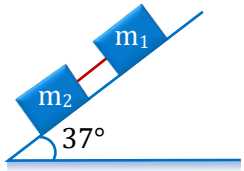
3. The minimum force required to start pushing a body up a rough (frictional coefficient μ) inclined plane is F_1 while the minimum force needed to prevent it from sliding down is F_2. If the inclined plane makes an angle θ from the horizontal such that $\tan\theta = 2\mu$ then the ratio $\frac{F_1}{F_2}$ is :-

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(A) 4 (B) 1 (C) 2 (D) 3

PNL316

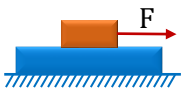
4. Two blocks $m_1 = 4\ kg$ and $m_2 = 2\ kg$, connected by a weightless rod on a plane having inclination of 37° as shown in figure. The coefficients of dynamic friction of m_1 and m_2 with the inclined plane are $\mu = 0.25$. Then the common acceleration of the two blocks and the tension in the rod are :



(A) $4\ m/s^2, T = 0$ (B) $2\ m/s^2, T = 5\ N$ (C) $10\ m/s^2, T = 10\ N$ (D) $15\ m/s^2, T = 9N$

PNL317

5. In the figure shown, the friction coefficient between the block of mass 1 kg and the plank of mass $2kg$ is 0.4 while that between the plank and floor is 0.1. A constant force 'F' starts acting horizontally on the upper 1 kg block. The acceleration of plank if $F = 10\ N$ is

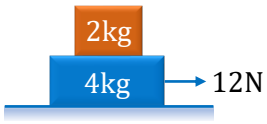
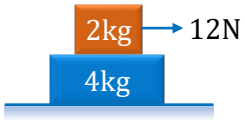
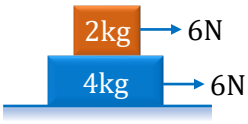


(A) $2.5\ m/s^2$ (B) $3.833\ m/s^2$ (C) $0.5\ m/s^2$ (D) $1.0\ m/s^2$

PNL318

For More Join: @IITJEE_Advanced

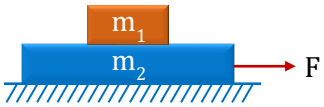
6. Column-II gives certain situations involving two blocks of mass 2 *kg* and 4 *kg*. The 4 *kg* block lies on a smooth horizontal table. There is sufficient friction between both the blocks and there is no relative motion between the blocks in all situation. Horizontal forces act on one or both blocks as shown. Column-I give certain statement related to figures given in column-II. Match the statements in column-I with the figure in column-II.

Column-I		**Column-II**
(A) Magnitude of frictional force is maximum.	(p)	
(B) Magnitude of friction force is least.	(q)	
(C) Friction force on 2 *kg* block is towards right.	(r)	
(D) Friction force on 2 *kg* block is towards left.	(s)	

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PNL319

7. A bar of mass m_1 is placed on a plank of mass m_2 which rests on a smooth horizontal plane. The coefficient of friction between the surfaces of bar and plank is k. The plank is subjected to a horizontal force F depending on time t as $F = at$, where a is a constant. The moment of time t_0 at which the plank starts sliding is :



(A) $\frac{akg}{m_1+m_2}$ (B) $\frac{(m_1+m_2)kg}{a}$ (C) $\frac{(m_1+m_2)g}{ka}$ (D) $\frac{ka}{(m_1+m_2)g}$

PNL320

8. In the given figure the coefficient of friction between 4 *kg* and 5 *kg* blocks is 0.2 and between 5 *kg* block and ground is 0.1. Choose the correct statements:

(A) Minimum force needed to cause system to move is 17 *N*.

(B) When force is 4 *N* static friction at all surfaces is 4 *N* to keep system at rest.

(C) Maximum acceleration of 4 *kg* block is 2 m/s^2.

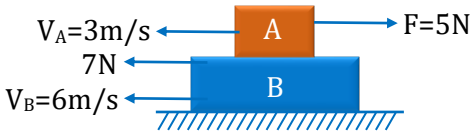
(D) Slipping between 4 *kg* and 5 *kg* blocks starts when *F* is > 17*N*.



PNL321

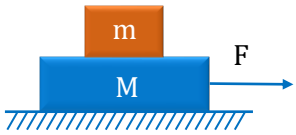
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9. In the following figure, find the direction of friction on the blocks and ground.



PNL329

10. Two blocks (m = 0.5 kg and M = 4.5 kg) are arranged on a horizontal frictionless table as shown in figure. The coefficient of static friction between the two blocks is $\frac{3}{7}$. Then the maximum horizontal force that can be applied on the larger block so that the blocks move together is _____ N.
(Round off to the Nearest Integer) [Take g as 9.8 ms^{-2}]



PNL323

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	3	D	A	C	(A-s), (B-r), (C-p, s), (D-q, r)	B	C		21

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